LEVELHULL · DESIGN DECISION RECORD

LevelHull concept selection

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DOCUMENT CONTROL

Document ID	LVH-DDR-004	Version	0.2
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/decisions/0004-concept-selection.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Option A cost USD 694 (four bailers, D-O3); the selection itself stays proposed, awaiting Amish (no pick)
0.1	2026-10-24	Amish Chadha	Concept selection argued (kit 1.9.1 upgrade); A and B within 10 %, both to Amish

0004: Concept selection

- **Date:** 2026-10-24
- **Status:** proposed. **Proposed, awaiting Amish.** No option is picked: options A and B scored within 10 % of each other, so both go to Amish (STANDARDS section 21). Until he decides, the repo stays on option A, its existing design.

> **Safety:** Whichever option is chosen, it is flotation for the boat and never a substitute for life jackets.

Context

LevelHull was designed before kit 1.8, so its existing design (foam modules high inside the hull, LVH-DDR-001 to LVH-DDR-003) is option A. Kit 1.8 asks for at least two concepts with different working principles, scored against the user requirements (LVH-URS-001). Two more were developed and modelled at D3 in `cad/src/model.py`: B, the same foam carried outside the hull as sponsons, and C, double outrigger floats. Their swamped performance was solved with the method of LVH-CAL-001 (`docs/04-calcs/stem_check.py`).

Options considered

Table 1. Summary (full scoring in LVH-OPT-001 Table 2).

	A: foam modules high inside	B: external foam sponsons	C: double outrigger floats
Weighted score (of 215)	172 (80 %)	163 (76 %)	148 (69 %)
Swamped lowest freeboard	92 mm	158 mm	160 mm
Roll stiffness	761 kg m per radian	1,546	6,313
Largest unit lost	Afloat, 25 mm	Gunwale 32 mm under	Capsizes
Inside volume taken	8.4 %	None	None
Beam	1.68 m	1.93 m	4.45 m
New hull holes	8 in the top plank	32 through the planks	16 through the strakes
Cost per canoe	USD 694	About USD 620 (estimate)	Pipe price to assess

Recommendation

Both A and B go to Amish:

- **A** if keeping the canoe's lines, the fewest hull holes and survival with one module lost matter most. It is already constructable, priced and planned.
- **B** if freeboard, roll stiffness and an empty inside matter most, accepting 32 bolt holes through the planks, a wider canoe, exposed covers and a redesign of the sponson split to meet R12.

C is not put forward: it scores lowest, a lost float capsizes the canoe, and its 4.45 m beam and booms conflict with landing and net handling; its strength, intact stability, needs the outriggers themselves and cannot be grafted.

Graft from the runner-up (for Amish with this decision). B shows that the same foam gives more freeboard when it sits lower while still cutting the waterline. Lowering A's modules and battens by about 40 mm (top 70 mm below the gunwale) would raise A's swamped freeboard at some cost in clear width and roll stiffness; it changes a design dimension, so it is offered as an option to the decision, not made.

Consequences

- `project.yaml` : `concept_options: [A, B, C]` ; `concept_chosen` left as "A (proposed)".
- LVH-DEC-001: open decision D-O1.
- The drawing, priced BOM and build plan stay on option A until Amish decides; if he picks B, those are redone for B and a design-for-construction record follows.